

Exam Date & Time: 17-Jun-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester Makeup Examination, June -2025

SOFT COMPUTING PARADIGMS [CSE- 4054]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A neural network receives inputs of (0.50, 0.75, 0.60) with weights (2.2, 1.8, 2.5) and a bias of 0.5. The activation function is sigmoid:
 $f(x) = 1 / (1 + e^{-x})$. If the actual output is 0.95, then: (5)
- A) i) Compute the net input and apply the activation function,
 ii) Draw the architecture of the neuron,
 iii) Suggest one method to minimize the error.
- B) A transportation agency is building a traffic congestion prediction system. Compare how hard and soft computing methods can be applied to this scenario. Discuss strengths and limitations in both cases. (5)

- 2) Using K-means clustering, categorize 4 smartphone models based on their battery capacity and average screen-on time per day. Perform one iteration of the algorithm and update the cluster centroids accordingly.
- A)

Smartphone	Battery capacity (mAh)	Screen time (Hours)
S ₁	3200	10
S ₂	3900	12
S ₃	4800	14
S ₄	6000	18
S ₅	2800	11

- B) Analyze the process of SOM training. How do they affect neuron adaptation and convergence? (5)
- 3) (5)
- A)

Consider a BPNN with:

Input Layer: 2 neurons with inputs [1.0,0.5]

Hidden Layer: 2 neurons with weights

$$w_{11} = 0.4, w_{12} = 0.3 \text{ (for neuron 1)}$$

$$w_{21} = 0.5, w_{22} = 0.6 \text{ (for neuron 2)}$$

Output Layer: 1 neuron with weights from hidden:

$$v_1 = 0.7, v_2 = 0.8$$

Bias: 0.1 for hidden and 0.2 for output

Activation: Sigmoid function

Tasks:

- i) Compute the output from each hidden neuron
- ii) Compute the final output from the network
- iii) If target = 0.9, calculate the error (use MSE)

- B) Analyze the flow of Mamdani and Sugeno fuzzy inference systems with an example involving room temperature control. (5)

- 4) A fuzzy logic system controls Fan Speed based on Temperature and Humidity as input variables. The fuzzy sets are defined as:

- A) Temperature: {Low, Medium, High}
 Humidity: {Dry, Normal, Wet}
 FanSpeed (Output): {Slow, Moderate, Fast}

The following rules are used:

IF Temperature is High AND Humidity is Dry
 THEN FanSpeed is Fast

IF Temperature is Medium AND Humidity is Normal THEN FanSpeed is Moderate

IF Temperature is Low OR Humidity is Wet THEN FanSpeed is Slow (5)

Given the input values:

Temperature = 78°F → $\mu_{\text{High}} = 0.7$, $\mu_{\text{Medium}} = 0.3$,
 $\mu_{\text{Low}} = 0$

Humidity = 40% → $\mu_{\text{Dry}} = 0.6$, $\mu_{\text{Normal}} = 0.4$, $\mu_{\text{Wet}} = 0$

Compute the firing strength for each rule using AND/OR operators (use min for AND, max for OR. Briefly explain how these will contribute to the final FanSpeed decision (no need to defuzzify)

- B) Given chromosomes: $x_1 = 4\ 3\ 2\ 1\ 5\ 6\ 7\ 8$ and $x_2 = 8\ 7\ 6\ 5\ 4\ 3\ 2$. Use fitness = count of even numbers. Rank and perform crossover at 5. (5)

- 5) How does ACO simulate the natural process of ant behavior? Analyze how pheromone updates and heuristic desirability influence the search. (5)
- A)
- B) Differentiate between Sugeno and Mamdani-based ANFIS architectures with suitable examples. (5)

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